

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 09 · LECTURE NOTES

Matching

Allocation without money, and the one place this theory is routinely deployed.

MODULE OUTCOMES

- State the stable matching problem and Gale-Shapley.
- Explain which side the algorithm favours and why.
- Explain strategy-proofness for one side only.
- Explain school choice and kidney exchange.
- State what a matching mechanism guarantees.

1 Stable matching

A MATCHING IS STABLE IF NO PAIR WOULD BOTH RATHER HAVE EACH OTHER THAN THEIR ASSIGNED PARTNERS

A **blocking pair** is two agents who each strictly prefer the other to their current assignment — and if such a pair exists they will simply arrange it privately, outside the mechanism, which destroys the matching and the mechanism's authority with it.

So **stability is not an aesthetic criterion** — it is precisely the condition under which the matching survives contact with the participants — which is why unstable mechanisms have historically unravelled, with participants matching earlier and earlier outside the process until it collapsed.

And Gale-Shapley (deferred acceptance) produces one: each unmatched agent on the proposing side proposes to its most preferred acceptable partner that has not yet rejected it; each agent on the receiving side holds its best offer so far and rejects the others, repeating until nobody is proposing.

It terminates in $O(n^2)$ and always produces a stable matching — so a stable matching always exists, which was not obvious and is the **theorem**: an existence proof that is also an algorithm, which is the opposite situation from Module 03.

2 Who it favours

THE PROPOSING SIDE GETS ITS BEST STABLE PARTNER AND THE RECEIVING SIDE ITS WORST

Among all stable matchings, deferred acceptance gives every agent on the proposing side the best partner it has in any stable matching — and every agent on the receiving side its worst — simultaneously, which is a strong and somewhat surprising result.

So which side proposes matters a great deal, and it is a policy choice rather than an implementation detail — which is exactly how the medical residency match was reformed when it was changed to be applicant-proposing.

And truthful reporting is a dominant strategy for the proposing side — and is not for the receiving side, which can sometimes benefit by truncating its preference list to make itself appear more selective.

Which is a specific and useful asymmetry: 'strategy-proof' here means strategy-proof for one side, and saying which side is part of describing the mechanism honestly (Module 13 §1's requirement). A paper or system that says only 'strategy-proof' has omitted half the claim.

3 Deployments

MEDICAL RESIDENCY MATCHING

the original large-scale deployment.
 Complication: couples wanting positions in the same city, which makes a stable matching possibly NONEXISTENT and the problem NP-hard.
 Handled in practice by a well-tested heuristic.

SCHOOL CHOICE

students to schools. Complication: schools are objects with priorities, not agents with preferences, so "stability" means respecting those priorities rather than mutual preference. The older Boston mechanism rewarded strategic ranking heavily -- families who ranked their true first choice could lose their second -- and deferred acceptance replaced it explicitly so that families would not need to game it.

KIDNEY EXCHANGE

incompatible donor-patient pairs swapped in cycles and chains. No money, by law.
 Complication: cycles must be short because all surgeries in a cycle happen simultaneously, which makes the optimisation NP-hard; chains started by altruistic donors relax the simultaneity requirement.

IN ALL THREE THE THEORY GOT DEPLOYED, which is rare, and in all three it needed real engineering.

In all three the theory was deployed and needed real engineering — which makes matching the field's best evidence that any of this works, and the best available answer to somebody who regards mechanism design as an elegant irrelevance. **School choice is the clearest case of a mechanism changed specifically to remove the incentive to game:** the reform's stated goal was that families should be able to report their true preferences safely, which is Module 01 §1's whole argument, enacted.

4 What it guarantees

- **It is stable — no pair can profitably deviate together** — which is what makes it survive, and is the property the whole construction is for.
- **It is not optimal in any aggregate sense — stability is a constraint, not an objective, and the set of stable matchings can differ a great deal in total welfare.** Nothing in the algorithm is maximising anything global.
- **It is not fair — the proposing side does systematically better (§2), by construction and not by accident** — which is why the choice of proposing side is a distributive decision.

- **And it is only as good as the reported preferences — which are truthful for one side and not necessarily for the other, and which people may not know accurately about themselves** in the first place (Module 01 §4's agent model).
- **Plus the preferences themselves are shaped by the mechanism: people rank differently when they believe the ranking matters strategically — which is precisely the effect the school choice reforms removed**, and which means comparing pre- and post-reform preference data is not comparing like with like. **Stability is a constraint, not an objective**, and saying otherwise is the standard misreading.

AND THIS IS THE FIELD'S BEST EVIDENCE

Matching is where mechanism design is routinely deployed and has demonstrably improved outcomes — residency matching, school choice, and kidney exchange, all operating at national scale and all with measurable results.

And in each case the deployment required handling something the clean theory excluded — couples, school priorities, cycle length limits — which is the engineering (Module 08 §4's pattern, in a second setting).

Which is the right note on which to assess this subject: the theory is not merely elegant, and it is also not sufficient on its own — both halves, and the second half is why the deployments took decades.

So when claiming a matching mechanism works: name the side it favours, the side it is strategy-proof for, and the complication you handled heuristically — three things, and Module 13 wants all three.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Gale & Shapley — College admissions and the stability of marriage (free)	§1 in the original — nine pages, and entirely readable.
Roth — Who Gets What and Why	§3 — the deployments described by the person who built several of them, including the failures. Library copy.
Abdulkadiroglu & Sönmez — School choice: a mechanism design approach (free)	§3's second case — the Boston mechanism's problem, and the reform argument.
Nisan et al., chapter 10 (free PDF)	§§1, 2, and 4 — the computational treatment, including the NP-hard variants.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Define stability** and give a blocking pair in a small instance.
- 2 **Run Gale-Shapley by hand** on a four-by-four instance.
- 3 **Run it with the other side proposing** and compare the outcomes.
- 4 **Enumerate all stable matchings** of a small instance and verify the extremal result.
- 5 **Find a profitable truncation** for a receiving-side agent.
- 6 **Implement deferred acceptance** and test it at scale.
- 7 **Add couples** and find an instance with no stable matching.
- 8 **Simulate the Boston mechanism** and find a family that is punished for honesty.
- 9 **Implement kidney exchange** with a cycle length cap, and measure what the cap costs.
- 10 **Write the three-part claim** for a matching mechanism you design.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define stability and a blocking pair.
- 2 Why is stability the right criterion rather than an aesthetic one?
- 3 Describe Gale-Shapley and state what it guarantees.
- 4 Which side does it favour, and how strongly?
- 5 For which side is it strategy-proof?
- 6 Name three deployments and the complication in each.
- 7 What did the school choice reform set out to remove?
- 8 Give four things a stable matching does not guarantee.
- 9 What is the standard misreading?
- 10 What three things should a matching claim name?